

# Manu Nicholas Jacob

Hardware Engineer, Root Cause Engineering · Dell Technologies · Austin, Texas

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Hardware engineer and independent researcher. Board-level failure analysis on enterprise AI server platforms, and 14 manuscripts, 9 under peer review at IEEE and ACM venues, on inference performance, power and memory behaviour.

## WORK EXPERIENCE

### Hardware Engineer · Dell Technologies, Root Cause Engineering, Austin, TX

Jun 2025 to present

Hardware engineer on enterprise AI server platforms: system-level and board-level debug, failure analysis and root-cause engineering across PCIe, memory, power and GPU subsystems, isolating faults by controlled reproduction rather than component swapping.

- Built an **agent-based** failure-analysis tracking system linking quality to engineering, centralizing case ownership and status and shortening turnaround in the rapid-response workflow.
- **Named inventor** on a Dell Technologies invention disclosure authorized for U.S. patent filing.
- On a **performance-analysis rotation** within the role (Feb to Apr 2026), built an automated server benchmarking and power-telemetry platform that replaced a 4-hour manual workflow with a 100-minute run, **62% faster** and 5 minutes of operator time.
- Diagnosed a riser detection failure down to a single open boost-converter inductor using an oscilloscope, logic analyzer and micro X-ray imaging, proving mechanical solder-joint separation rather than electrical overstress, and separated genuine component failures from system-level interactions across recurring controller errors and repeat no-boot cases.

### Root Cause Engineering Intern · Dell Technologies, Austin, TX

May to Aug 2024

- Developed a unified diagnostic tool for NVIDIA AI platforms that detected PCIe replays, bus resets and GPU faults through a single workflow.
- Implemented optimized secondary-bus-reset procedures and real-time PCIe replay detection, cutting link retrain time and improving fault isolation on the debug bench.

### Controls Engineering Intern · Schneider Electric, Lexington, KY

Jun to Aug 2023

- Designed and deployed a human-machine interface for palletizer control on the factory floor and optimized PLC-to-HMI communication on the packer line, improving line efficiency and reducing downtime.
- Led an overhaul of internal standardization specifications against ANSI/RIA and OSHA requirements, adding a dedicated section for robotic cells.

## SKILLS

**Hardware:** enterprise servers, PCIe, GPU subsystems, DDR memory, power delivery, board-level debug, oscilloscope, logic analyzer, X-ray inspection, embedded systems, microcontrollers, custom PCB.

**Software:** Python, C, C++, SQL, Bash, Linux, PyTorch, llama.cpp, Linux perf, hardware performance counters.

**Infrastructure:** Kubernetes (k3s), Docker, cgroups, Prometheus, Grafana, distributed job scheduling.

**Certifications:** NVIDIA AI Infrastructure and Operations Fundamentals (2026), and 3 Oracle Cloud Infrastructure 2025 certifications: Generative AI Professional, AI Foundations, Foundations.

## EDUCATION

### University of Massachusetts Amherst · B.S. Electrical Engineering and B.S. Computer Engineering

2021 to May 2025

## RESEARCH AND PUBLICATIONS

- **14 manuscripts, 9 under peer review** at IEEE Transactions on Computers, ACM TECS, IEEE Internet of Things Journal, IEEE Embedded Systems Letters, IEEE Design & Test and HotMobile 2027.
- 4 TechRxiv preprints; every result ships with a DOI-archived artifact. manunicholasjacob.com/research.

## OPEN SOURCE AND PROFESSIONAL SERVICE

Merged 3 of 11 submitted pull requests into **NVIDIA garak**, authored 4 open pull requests to **vLLM**; published **llama-roofline** and **ML Systems Lab** to PyPI, plus **8 DOI-archived research artifacts** on Zenodo. **Artifact Evaluation Committee**, ATC '26 (ACM SIGOPS, formerly USENIX ATC; CORE A); **reviewer**, **Journal of Open Source Software** (one review completed, three assigned); **Judge**, **HackTX 2025** (UT Austin, 700+ participants).